

Assignment 01 — Observability in the Wild

Adding OpenTelemetry to nopCommerce

Sérgio Correia - 108647

The Problem

nopCommerce

A production e-commerce platform with no telemetry.

The goal: pick one user flow and instrument it end-to-end.

User Flow

Customer places an order

How Observability Entered the System

OTelStartup

Auto-discovered at startup — no existing files modified

- OTel SDK setup in OTelStartup.cs
- No existing classes modified

NopActivitySource

Placed in Nop.Core

- Centralizes ActivitySource and Meter
- No OTel SDK dependency in domain
- Separation of concerns preserved

Spans Created



checkout.place_order

CheckoutController - total checkout duration, payment method, orderID, storeID



order.prepare

OrderProcessingService - validation phase duration



order.place

OrderProcessingService - cart items, order total, currency



payment.result

Payment gateway call - success/error



db.insert

EntityRepository - All writes to db, entity type only

Privacy Strategy

Where

SDK level - data never enters the telemetry pipeline

Excluded from Spans

- Emails, names, addresses, card data, phone number, IP address

What's included

- Payment method, store ID, order ID, entity type, order total, currency, cart items

Why at SDK level

Don't put PII in spans. Redacting it later is damage control.

Custom Metrics

1

checkout.cart_age_seconds

Cart lifetime until checkout. Values below 15s indicate automated traffic (bots)

2

checkout.db_write_duration

SaveOrderDetailsAsync duration in ms. Value rising isolates database degradation

3

checkout.provider_outcome

Payment attempts by result: success / timeout. Isolates payment provider failures

Not Instrumented

IEventPublisher

Ideal instrumentation boundary - every domain event passes through it. Can't be decorated due to internal static service locator. Would require modifying the publisher itself.

Credit card decline metric

Wanted to detect carding attacks via abnormal decline rates. The payment method never declines - accepts any valid card format. Replaced with timeout simulation via `PAYMENT_FAILURE_RATE`.

DEMO

Grafana Dashboard & Load Test